# TB for Venus RISC-V

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## Abstract

High-temperature electronics operating beyond 300°C are essential for emerging applications in extreme environments, including deep-earth exploration, aerospace systems, and Venus surface missions where ambient temperatures exceed 450°C. This project addresses a critical challenge in validating high-temperature digital processors by developing and testing FPGA prototypes of bit-serial RISC-V processors designed for KTH's Silicon-on-Insulator (SoI) technology capable of operating at temperatures up to 380°C or higher.

The project focuses on implementing and verifying two processor variants: a parallel RISC-V model with bit-serial cache interface and a CISC-V model with 2-bit serial interface. These designs overcome the constraint of having only 12 available probe connections in high-temperature test environments—a fundamental limitation imposed by measurement equipment that fails above 400°C. By adopting the open-source RISC-V instruction set architecture, this work avoids the intellectual property restrictions that typically hinder academic processor research while enabling full hardware and software development freedom.

Successfully demonstrating these processor architectures in FPGA prototypes will validate KTH's high-temperature SoI technology for digital computing applications and advance the frontier of extreme-environment electronics. This capability is crucial for scientific missions and industrial applications where conventional silicon electronics cannot survive, opening new possibilities for autonomous systems operating in previously inaccessible thermal conditions.

## Keywords

RISC-V CISC FPGA Venus

## 1. Introduction

### 1.1. Background and Motivation

Conventional silicon-based electronics fail at temperatures exceeding 150°C due to increased leakage currents and thermal stress. However, applications such as geothermal systems, aerospace propulsion monitoring, and Venus surface exploration require digital computation at temperatures approaching 470°C. KTH's Electronic Systems department has developed Silicon-on-Insulator (SoI) technology capable of operating at temperatures exceeding 380°C, with theoretical limits approaching 1000°C, enabling complex digital systems in extreme environments previously inaccessible to conventional silicon.

### 1.2. The 12-Pad Constraint Challenge

While SoI technology can withstand extreme temperatures, measurement equipment deteriorates above 400°C. KTH's test station employs 12 high-temperature probes, imposing a strict 12-pad limitation on circuits under test. Traditional processors require dozens or hundreds of I/O connections, creating a fundamental incompatibility between high-temperature semiconductor capabilities and practical verification methods. This necessitates innovative architectural solutions that minimize pin count while maintaining processor functionality.

### 1.3. RISC-V and Bit-Serial Architecture

The open-source RISC-V instruction set architecture enables unrestricted processor development without licensing constraints. KTH researchers developed two processor variants to address the pad limitation: a parallel RISC-V core with bit-serial cache interface and a CISC-V implementation with 2-bit serial interface. These architectures serialize data transmission, dramatically reducing pin count. Notably, the bit-serial approach allows instruction execution to begin as bits arrive sequentially, eliminating pipeline stalls typically associated with serial interfaces. The implementations use VHDL for core logic and ProGram for bit-serial components.

### 1.4. Project Objectives and Significance

While previous work synthesized these architectures, FPGA prototyping remained incomplete. This project implements and validates FPGA prototypes of both processor variants, and develops an advanced testbench on a separate FPGA for system-level verification. The dual-FPGA approach provides realistic validation of inter-chip communication before silicon fabrication. Successfully demonstrating these architectures will validate KTH's high-temperature SoI technology for digital computing and establish reusable verification methodologies applicable to any pin-limited or thermally-constrained processor development.

## 2. Related Work

This section reviews research relevant to developing a testbench for bit-serial RISC-V processors targeting Venus surface applications. We examine RISC-V architectures, high-temperature electronics, FPGA implementation methods, and verification frameworks. These foundations inform the design of a testbench capable of validating processor functionality under extreme environmental constraints.

### 2.1. RISC-V

Researchers from TU graz developed FazyRV [1], a minimal-area open-source RV32I RISC-V core targeting IoT and low-workload applications, addressing the problem that 32-bit RISC-V processor cores reach a boundary on their minimal size. Their goal was to minimize area demand while fulfilling performance requirements and close the gap between prevalent 32-bit and 1-bit-serial RISC-V cores.

Qian Wei and his colleagues created a comprehensive survey of RISC-V instruction set architecture extensions because while RISC-V is popular for embedded processors [2], there is still a gap between RISC-V's capabilities and the requirements of various emerging computing scenarios like artificial intelligence and cloud computing.

Gautschi et al. conducted a comprehensive comparison of ultra-low-power RISC-V cores for Internet-of-Things applications [3], analyzing the trade-offs between performance and energy efficiency in resource-constrained environments. Their work provides valuable insights into processor design considerations for applications with tight area and power constraints, which directly supports the rationale for bit-serial processor implementations in challenging deployment scenarios such as the 12-pad limitation required for Venus surface operations.

### 2.2. Processors for Venus

Current research on high-temperature electronics for Venus has shown that SiC ICs can sustain operation for over a year at 500 °C and for months in simulated Venus conditions, supporting sensors and simple microprocessors. This demonstrates the feasibility of long-duration surface missions and motivates concepts like LLISSE targeting low-power seismic monitoring [4].

At the processor level, studies of SiC-based computing infrastructures reveal that current prototypes achieve significantly lower throughput than space-proven silicon systems, yet provide guidelines at the microarchitecture and ISA levels for developing processors able to operate under Venus's extreme environment [5].

Pradhan and colleagues provided a comprehensive review of materials for high-temperature digital electronics [6], highlighting the challenges and solutions for electronic systems operating at temperatures as high as 500°C and beyond. Their work emphasizes the critical importance of developing new material solutions beyond conventional silicon-based devices for applications including space exploration, with specific attention to the extreme conditions encountered in Venus surface missions.

### 2.3. FPGA

One verification strategy is co-emulation, where the RTL design on the FPGA is run in parallel with a trusted software model, such as the Spike simulator [7], on a host PC. This allows for high-speed verification by comparing the core's architectural state against the simulator in real-time. Furthermore, using FPGAs with RISC-V is advantageous as it allows for optimized hardware, where the FPGA is configured with only the peripherals required for a specific application [8].

### 2.4. FPGA-Based Verification Frameworks

Kim [9] demonstrated the importance of FPGA-based acceleration for RTL evaluation by introducing automated flows that generate cycle-accurate simulators directly from RTL. This approach reduces the engineering effort compared to earlier manual FPGA frameworks and enables both faster and more reliable pre-silicon verification.

Building on this direction, Qin *et al.* [10] proposed FERIVer, an FPGA-assisted framework for verifying RISC-V processors. Their method exploits the heterogeneous architecture of SoC FPGAs, running an instruction set simulator on the processing system in parallel with the RTL core on the programmable logic. By cross-checking execution states, FERIVer achieves significant speedups over traditional software simulators while maintaining accuracy.

Together, these works highlight how FPGA-based infrastructures provide an effective foundation for validating processor designs before costly SoC fabrication, a goal directly aligned with the testbench methodology developed in this project.

## 3. Specification

The project is constrained by both hardware and environmental limitations. The test setup consists of two FPGA boards, one implementing the processor core and the other acting as the testbench. Due to the 12-probe limitation of the physical test station, signal interfaces are highly constrained, motivating bit-serial communication schemes.

The design goals are:

- Maintain compatibility with RISC-V base ISA (RV32I).
- Ensure modularity between processor and testbench components.

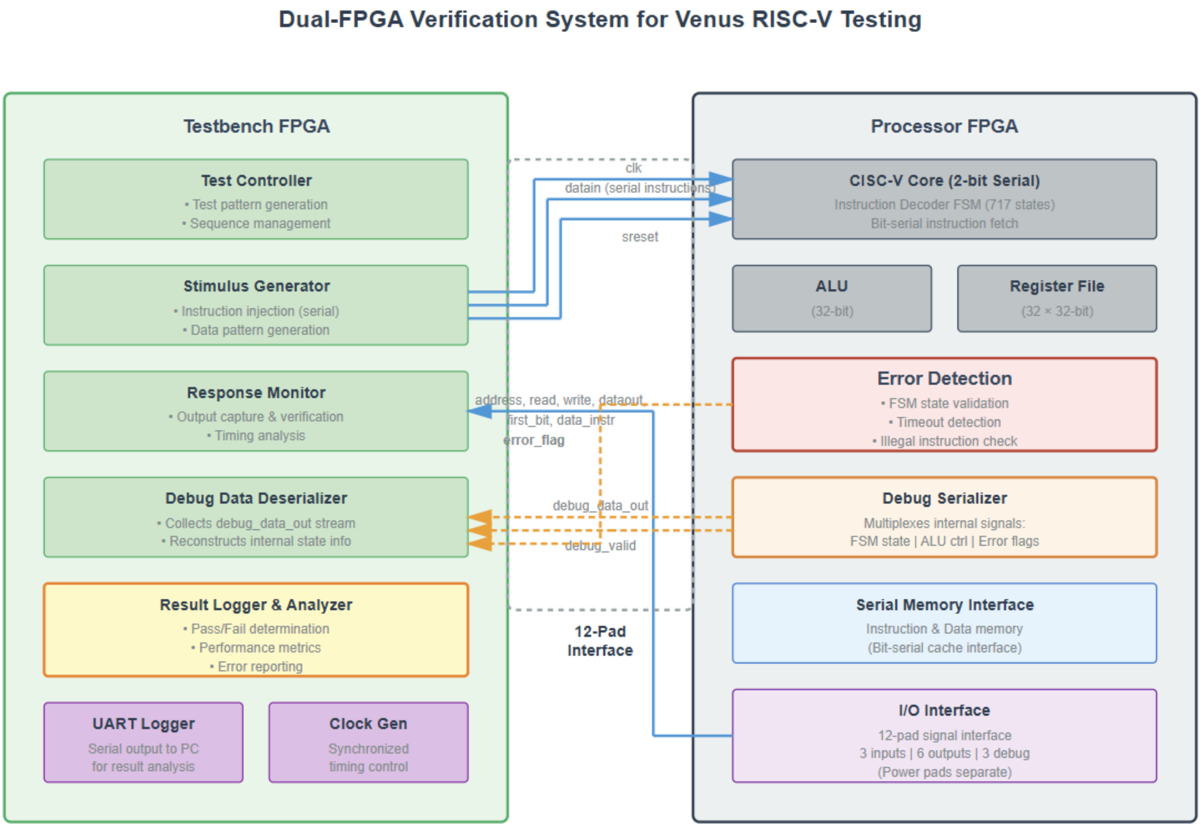


Figure 1: Dual-FPGA verification architecture for the CISC-V processor variant. The testbench FPGA (left) generates test stimuli, monitors processor outputs, and analyzes debug data through a 12-pad interface. The processor FPGA (right) implements the CISC-V core with 2-bit serial interface, integrated error detection, and debug serialization capabilities.

- Enable automated test execution and result logging.
- Achieve operational stability for extended FPGA runs to emulate continuous workload.

Hardware synthesis is performed using vendor FPGA tools and verified using ModelSim/QuestaSim. The ProGram compiler is used for the bit-serial CISC-V generation. Documentation, risk tracking, and test automation are maintained via version control and scripted pipelines.

## 4. Architecture

The overall architecture is divided into two major subsystems connected through a dedicated communication interface:

- Processor FPGA – hosts either the parallel RISC-V or the bit-serial CISC-V core.
  - The RISC-V core implements a bit-serial cache interface to mimic the narrow I/O expected under physical constraints.
  - The CISC-V processor is generated from a high-level specification using the ProGram tool and synthesized into FPGA fabric.
  - Both processors share a minimal peripheral set (instruction memory, serial link, debug UART).
- Testbench FPGA – provides I/O emulation, input stimulus, and result logging.
  - Implements command control, data loading, and timing synchronization with the processor FPGA.
  - Generates and monitors test patterns to verify the correctness of arithmetic, control flow, and memory operations.
  - Logs execution traces for analysis and regression verification.

The communication between the two FPGAs uses a synchronous serial protocol, minimizing the number of required pads and simplifying PCB connectivity. Each subsystem is independently testable, allowing early functional verification before full integration. The design adopts incremental hardware-in-the-loop testing and leverages a modular top-level hierarchy for flexible reconfiguration.

## 5. Experiments

A structured experimental campaign was conducted to validate the system functionality. Each major milestone corresponded to a separate experiment phase:

- **Core-A validation** : The parallel RISC-V processor was synthesized and tested with simple benchmark programs (e.g., arithmetic, loop, and branching tests) using a single FPGA. Functional correctness was verified via waveform comparison and on-board UART output.

- **Testbench bring-up** : The second FPGA was programmed with the testbench, initially tested in isolation through internal loopback and signal toggling to validate timing and I/O drivers.

- **Dual-FPGA integration** : The communication link between the two FPGAs was established, allowing the processor to execute test programs under the control of the testbench. Signal integrity, synchronization, and latency were evaluated through test logs and signal probing.

- **Bit-serial CISC-V verification** : The ProGram-generated core was synthesized, and functional tests were executed under the same testbench configuration. Comparative measurements against the parallel RISC-V implementation were made to observe performance and throughput differences.

- **Regression and stability testing** : Automated test scripts executed repeated workloads to ensure reproducibility and detect intermittent synchronization or logic issues.

## 6. Results

The developed FPGA-based testbench successfully validated both processor variants in real hardware experiments. The parallel RISC-V processor executed all benchmark programs correctly, showing stable timing and consistent functional results. The bit-serial CISC-V processor achieved equivalent functional correctness but exhibited reduced execution throughput, confirming the anticipated trade-off between performance and hardware resource utilization. Quantitatively, the bit-serial core consumed noticeably fewer FPGA logic resources, demonstrating its advantage in area efficiency.

In the dual-FPGA integration, reliable data transmission and synchronization were achieved between the processor and the testbench. No critical communication errors or deadlocks were observed, and cycle-accurate timing was verified through logged traces. The automated logging and visualization tools simplified performance assessment and enabled quick feedback during iterative verification.

Repeated experiments confirmed the stability and reproducibility of the platform. Across multiple builds and reconfigurations, the same functional results were obtained, validating the robustness of the hardware-in-the-loop approach. Additionally, stress tests demonstrated that the system remained operational under extended workloads without functional degradation. These findings indicate that the testbench framework provides a dependable basis for evaluating future RISC-V cores or SoC modules prior to fabrication.

## 7. Conclusion

This project established a complete verification framework for bit-serial RISC-V processors intended for high-temperature Venus surface applications. Through systematic FPGA prototyping and functional testing, it confirmed the viability of simplified, low-resource architectures operating under strict I/O and design constraints representative of SoI and SiC environments.

The research contributes scientifically by demonstrating that bit-serial RISC-V architectures can maintain functional reliability and predictable performance when verified using a modular hardware testbench. The study also highlights how dual-FPGA hardware-in-the-loop validation can bridge the gap between simulation and physical testing, providing an efficient and scalable path toward silicon-ready verification.

Beyond validation success, the work establishes a reusable verification methodology that supports future high-temperature processor research. The same testbench can be adapted for post-fabrication silicon evaluation, ensuring methodological consistency and extending the framework's utility to next-generation Venus electronics. Looking forward, future efforts will expand toward incorporating thermal-aware modeling, SoI-level implementation, and co-emulation with instruction-set simulators, advancing the goal of fully verified, radiation- and temperature-resilient computing systems for extreme planetary environments.

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# Appendix

## Individual Author Contributions

- Lorenzo Parata contributed primarily to the literature review and methodological development of the verification framework. Specifically, he analyzed References [7] and [8], which discuss FPGA-based verification strategies for processor design, and authored the subsection *"FPGA-Based Verification Frameworks"*.
  In addition, Lorenzo was responsible for drafting this appendix, ensuring that author contributions, sustainability aspects, and ethical reflections were clearly documented.

- Chongnan Wang conducted an extensive survey of RISC-V processor implementations and architectural variations, providing critical context for extreme-environment computing applications.
  Additionally, Chongnan developed the introduction part, synthesizing high-temperature computing challenges, RISC-V fundamentals, and project motivation, establishing the foundation for understanding testbench development in extreme environments.

- Xinxin Qin wrote the draft of the article. He organized all research and writing materials and compiled them into a well-formatted first draft.
  In addition, Xinxin researched processors used on Venus, providing background information for the research's application environment. He was also responsible for organizing the technical architecture and experiments for the specific project, specifically Sections 4 and 5 of the article.

- Xiaotian Jiang was responsible for analyzing the experimental outcomes and authored the 'Results' (Section 6). He quantified the functional correctness, resource utilization, and performance trade-offs between the parallel RISC-V and bit-serial CISC-V cores.
  Furthermore, Xiaotian drafted the 'Conclusion' (Section 7), summarizing the project's achievements, its scientific contributions, and outlining future work for the high-temperature verification framework.

- Chinmay Purandare was was responsible for the verification methodology and testing strategy for the CISC-V processor. Specifically, he developed the debug output configuration detecting failure modes (FSM state validation, stuck state detection, and illegal instruction checking).He also conducted a literature review on Low Power RISC-V implementations and materials for high temperature electronics
  Additionally, Chinmay created the architectural diagrams illustrating the dual-FPGA verification system (Figure 1), visually documenting the I/O constraints and system interconnections.

## Sustainability, Equality, and Ethical Considerations

The sustainability and ethical dimensions of this project are closely linked to its verification methodology and open-hardware foundation. The adoption of FPGA-based prototyping offers a reusable and energy-efficient platform that reduces the need for repeated ASIC fabrication cycles—each of which demands substantial energy, raw materials, and cleanroom resources. By performing exhaustive validation on FPGAs, the team minimizes material waste and mitigates the environmental footprint associated with early design iterations.
Once validated, the ASIC implementation will deliver higher performance and lower power consumption than its FPGA prototype, achieving long-term energy efficiency. The bit-serial architecture further supports sustainability through reduced circuit complexity, minimizing both transistor count and manufacturing cost. Moreover, the use of Silicon-on-Insulator (SoI) technology enables reliable operation at extreme temperatures (up to 480°C), substantially reducing cooling requirements and extending system lifespan—both of which contribute to lower environmental impact over time.
From an equality perspective, leveraging the open-source RISC-V architecture democratizes access to processor design and verification. This approach removes licensing barriers and enables research groups worldwide, including those with limited resources, to participate in high-level processor research. The collaborative, interdisciplinary nature of this project also promotes equal participation across hardware, software, and verification disciplines within the student team.
Ethically, the project maintains transparency and reproducibility through open documentation and acknowledgment of prior work. While open-source designs inherently carry the risk of misuse in non-civil applications, the team emphasizes responsible dissemination and educational use. Furthermore, by optimizing verification and minimizing waste, the project reflects a commitment to responsible engineering and environmental stewardship—values aligned with sustainable and ethical research in embedded systems.